

How to Install Docker on Raspberry Pi

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DEBIAN DOCKER RASPBIAN

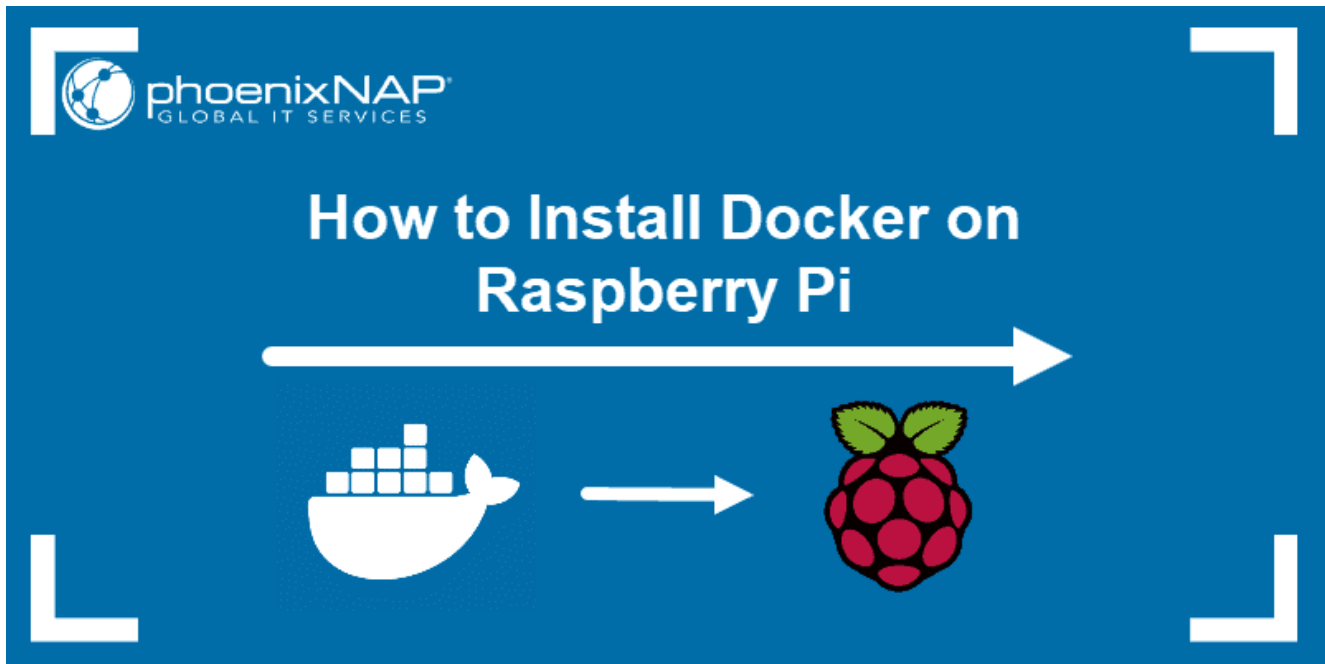
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Introduction

Docker is a tool for creating, deploying, and running applications in containers. The software is popular among developers as it speeds up the development process and does not use a lot of resources.

Docker containers are lightweight, especially compared to virtual machines. This feature is especially valuable if you are a Raspberry Pi user.

If you need help installing Docker on your Raspberry Pi, read our step-by-step guide on **how to install Docker on Raspberry Pi**.



Prerequisites

- Raspberry Pi with a running Raspbian OS
- Raspbian Stretch (Lite)
- SSH connection enabled

How to Install Docker on Raspberry Pi

To install Docker on your Raspberry Pi, you need to go through the following steps:

1. Update and upgrade your system.
2. Download the installation script and install the package.
3. Allow a non-root user to execute [Docker commands](#).
4. Verify installation by checking the Docker version.
5. Test the set up by running a “hello-world” container.

Step 1: Update and Upgrade

Start by updating and upgrading the system. This ensures you install the latest version of the software.

Open a terminal window and run the command:

```
sudo apt-get update && sudo apt-get upgrade
```



```
pi@osboxes:~$ sudo apt-get update
Ign:1 http://ftp.debian.org/debian stretch InRelease
Get:2 http://ftp.debian.org/debian stretch-updates InRelease [91.0 kB]
Get:3 http://archive.raspberrypi.org/debian stretch InRelease [25.4 kB]
Get:4 http://ftp.debian.org/debian stretch-backports InRelease [91.8 kB]
Get:5 http://security.debian.org stretch/updates InRelease [94.3 kB]
Hit:6 http://ftp.debian.org/debian stretch Release
Get:7 http://ftp.debian.org/debian stretch-updates/main amd64 Packages [27.9 kB]
Get:8 http://ftp.debian.org/debian stretch-updates/main i386 Packages [27.9 kB]
Get:9 http://ftp.debian.org/debian stretch-updates/main Translation-en [11.9 kB]
Get:10 http://archive.raspberrypi.org/debian stretch/main i386 Packages [127 kB]
Get:11 http://ftp.debian.org/debian stretch-backports/main amd64 Packages [607 kB]
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Get:19 http://security.debian.org stretch/updates/main Translation-en [224 kB]
Get:20 http://security.debian.org stretch/updates/non-free amd64 Packages [1,596 B]
Get:21 http://security.debian.org stretch/updates/non-free i386 Packages [1,596 B]
Fetched 3,583 kB in 9s (376 kB/s)
Reading package lists... Done
pi@osboxes:~$ sudo apt-get upgrade
Reading package lists... Done
Building dependency tree
Reading state information... Done
Calculating upgrade... Done
The following packages have been kept back:
  lxplug-ejecter lxplug-network
The following packages will be upgraded:
  file intel-microcode libarchive13 libgs9 libgs9-common libmagic-mgc libmagic1 linux-compiler-gcc-6
  linux-image-4.9.0-11-686-pae linux-kbuild-4.9 linux-libc-dev python-piglow python3-piglow
15 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
Need to get 59.8 MB of archives.
After this operation, 493 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
```



Note: To avoid security and performance issues make sure to [update Raspberry Pi](#) frequently.

Step 2: Download the Convenience Script and Install Docker on Raspberry Pi

Move on to downloading the installation script with:

```
curl -fsSL https://get.docker.com -o get-docker.sh
```

Execute the script using the command:

```
sudo sh get-docker.sh
```

This installs the required packages for your Raspbian Linux distribution.

```
pi@osboxes:~$ curl -fsSL https://get.docker.com -o get-docker.sh
pi@osboxes:~$ sudo sh get-docker.sh
# Executing docker install script, commit: f45d7c11389849ff46a6b4d94e0dd1ffebca32c1
+ sh -c apt-get update -qq >/dev/null
```

The output will tell you which version of Docker is now running on your system.

```
Server: Docker Engine - Community
Engine:
 Version:           19.03.5
 API version:       1.40 (minimum version 1.12)
 Go version:        go1.12.12
 Git commit:        633a0ea838
 Built:             Wed Nov 13 07:28:22 2019
 OS/Arch:           linux/amd64
 Experimental:      false
containerd:
 Version:           1.2.10
 GitCommit:         b34a5c8af56e510852c35414db4c1f4fa6172339
runc:
 Version:           1.0.0-rc8+dev
 GitCommit:         3e425f80a8c931f88e6d94a8c831b9d5aa481657
docker-init:
 Version:           0.18.0
 GitCommit:         fec3683
If you would like to use Docker as a non-root user, you should now consider
adding your user to the "docker" group with something like:

    sudo usermod -aG docker your-user

Remember that you will have to log out and back in for this to take effect!

WARNING: Adding a user to the "docker" group will grant the ability to run
containers which can be used to obtain root privileges on the
docker host.
Refer to https://docs.docker.com/engine/security/security/#docker-daemon-attac
k-surface
for more information.
```

Step 3: Add a Non-Root User to the Docker Group

By default, only users who have administrative privileges (root users) can run containers. If you are not logged in as the root, one option is to use the **sudo** prefix.

However, you could also add your non-root user to the Docker group which will allow it to execute docker commands.

The syntax for adding users to the Docker group is:

```
sudo usermod -aG docker [user_name]
```



To add the **Pi** user (the default user in Raspbian), use the command:

```
sudo usermod -aG docker Pi
```



There is no specific output if the process is successful. For the changes to take place, you need to log out and then back in.

Step 4: Check Docker Version and Info

Check the version of Docker on your Raspberry Pi by typing:

```
docker version
```



The output will display the Docker version along with some additional information. For system-wide information (including the kernel version, number of containers and images, and more extended description) run:

```
docker info
```



Step 5: Run Hello World Container

The best way to test whether Docker has been set up correctly is to run the *Hello World* container.

To do so, type in the following command:

```
docker run hello-world
```



The software will contact the Docker daemon, pull the “hello-world” image, and create a new container based on that image.

Once it goes through all the steps, the output should inform you that ***your installation appears to be working correctly.***

```
pi@osboxes:~$ sudo docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

pi@osboxes:~$
```



Note: Try out [creating containers on your own](#) and get acquainted with some of the [best practices for managing Docker containers](#).

Raspberry Pi Docker Images

Opposed to most other Linux distributions, Raspberry Pi is based on ARM architecture. Hence, not all Docker images will work on your Raspberry Pi.

In fact, all Docker images that are not specifically designed for ARM devices will not work. Running docker-apps designed for x86/x64 and i386 architecture on your Raspberry Pi will return an error.

Remember that when searching for images to pull from [Docker Hub](#). Apply the **Architectures** filter to search for supported apps.

For custom solutions, the best thing to do would be to run a container based on an official image and then [modify it and commit the changes](#) to a new image.

How to Upgrade Docker on Raspberry Pi?

There is no need to re-run the convenience script to upgrade Docker. Furthermore, running the convenience script again might cause issues if it attempts to repositories that have been added already.

Upgrade Docker using the package manager with the command:

```
sudo apt-get upgrade
```



How to Uninstall Docker on Your Raspberry Pi?

Although you used a convenience script for installation, you can simply remove docker using the package manager:

```
sudo apt-get purge docker-ce
```



Note: Depending on the version of software, you may need to use an additional command to completely remove Docker: **sudo apt-get purge docker-ce-cli.**

To delete leftover images, containers, volumes and other related data, run the following command:

```
sudo rm -rf /var/lib/docker
```



Edited configuration files must be deleted manually.

Conclusion

You are now set to use Docker on your Raspberry Pi and develop isolated and lightweight applications using containers.

Was this article helpful?

Yes

No

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Sofija Simic is an aspiring Technical Writer at phoenixNAP. Alongside her educational background in teaching and writing, she has had a lifelong passion for information technology. She is committed to unscrambling confusing IT concepts and streamlining intricate software installations.

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